

MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS
NANYANG TECHNOLOGICAL UNIVERSITY

Tutorial 10

HINTS

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 We will discuss several questions from Chapter 8 in Ross' book.

Tutorial Questions

- (1) Let $X_1, \dots, X_{20}$ be independent Poisson random variables with mean 1.

- (a) Use the Markov's inequality to obtain a bound on

$$\Pr \left\{ \sum_{i=1}^{20} X_i > 15 \right\}.$$

- (b) Use the central limit theorem to approximate the same probability in part (a).

Solution:

- (a) By Markov's inequality,

$$P \left\{ \sum_{i=1}^{20} X_i > 15 \right\} = P \left\{ \sum_{i=1}^{20} X_i \geq 16 \right\} \leq \frac{\mathbb{E} \left(\sum_{i=1}^{20} X_i \right)}{16} = \frac{\sum_{i=1}^{20} \mathbb{E}(X_i)}{16} = \frac{20}{16} = \frac{5}{4},$$

which is a trivial bound (it says a probability is less than some number that is greater than 1, which trivially holds).

- (b) Let $X = \sum_{i=1}^{20} X_i$, then $\mathbb{E}(X) = 20$ and $\text{Var}(X) = \sum_{i=1}^{20} \text{Var}(X_i) = 20$. By the central limit theorem, $(X - 20)/\sqrt{20}$ is approximately standard normal, and thus

$$\begin{aligned} P \left\{ \sum_{i=1}^{20} X_i > 15 \right\} &= P \left\{ \sum_{i=1}^{20} X_i \geq 15.5 \right\} = P \left\{ \frac{\sum_{i=1}^{20} X_i - 20}{\sqrt{20}} \geq \frac{15.5 - 20}{\sqrt{20}} \right\} \\ &= P \left\{ Z \geq -\frac{4.5}{\sqrt{20}} \right\} \approx P \{ Z \geq -1.006 \} = P \{ Z < 1.006 \} \approx 0.8428. \end{aligned}$$

- (2) Let $X_1, \dots, X_n$ be independent Poisson random variables with mean 1. Let $X = \sum_{i=1}^n X_i$.

- (a) Use the Markov inequality to obtain a bound on

$$P \left\{ \sum_{i=1}^n X_i \geq \frac{3}{2}n \right\}.$$

- (b) Use Chebyshev's inequality to obtain a bound on the same probability in part (a).
 (c) Derive a bound on the same probability in part (a) by considering

$$P\{|X - \mathbb{E}(X)| \geq t\} = P\{|X - \mathbb{E}(X)|^4 \geq t^4\} \leq \frac{\mathbb{E}(|X - \mathbb{E} X|^4)}{t^4}.$$

You may want to use a results from Tutorial 4.

(d) (optional) Derive a bound on the same probability in part (a) by considering

$$P\{X \geq t\} = P\{e^{\theta X} \geq e^{\theta t}\} \leq \frac{\mathbb{E}(e^{\theta X})}{e^{\theta t}}, \quad \theta > 0$$

and taking $\theta = \ln(t/n)$.

You should see better and better upper bounds (better means smaller) from part (a) to part (d) for large n . This is a general phenomenon, not restricted to the sum of Poisson random variables. In general, in part (d), instead of taking θ to be $\ln(t/n)$ one can choose θ to minimize the upper bound $e^{-\theta t} \mathbb{E}(e^{\theta X})$.

Solution: Note that $X \sim \text{Poisson}(n)$, $\mathbb{E}(X) = n$ and $\text{Var}(X) = n$.

(a) By Markov's inequality,

$$P\left\{X \geq \frac{3}{2}n\right\} \leq \frac{\mathbb{E}(X)}{\frac{3}{2}n} = \frac{2}{3}.$$

(b) By Chebyshev's inequality,

$$P\left\{X \geq \frac{3}{2}n\right\} = P\left\{X - \mathbb{E}(X) \geq \frac{1}{2}n\right\} \leq P\left\{|X - \mathbb{E}(X)| \geq \frac{1}{2}n\right\} \leq \frac{\text{Var}(X)}{\frac{1}{4}n^2} = \frac{4}{n}.$$

(c) First,

$$P\{|X - n| \geq t\} = P\{|X - n|^4 \geq t^4\} \leq \frac{\mathbb{E}(|X - n|^4)}{t^4}.$$

Note that

$$\mathbb{E}(X - n)^4 = \mathbb{E}(X^4) - 4 \cdot \mathbb{E}(X^3) \cdot n + 6 \cdot \mathbb{E}(X^2) \cdot n^2 - 4 \cdot \mathbb{E}(X) \cdot n^3 + n^4,$$

and thus we shall need to calculate $\mathbb{E}(X^2)$, $\mathbb{E}(X^3)$ and $\mathbb{E}(X^4)$. The second moment is easy,

$$\mathbb{E}(X^2) = \text{Var}(X) + (\mathbb{E} X)^2 = n + n^2.$$

Since $X \sim \text{Poisson}(n)$, we have the following

$$\mathbb{E}(X^k) = n \mathbb{E}((X + 1)^k).$$

Check it:

$$\begin{aligned} \mathbb{E}(X^k) &= \sum_{j=1}^{\infty} j^k \left(e^{-n} \frac{n^j}{j!} \right) = \sum_{j=0}^{\infty} (j+1)^k e^{-n} \frac{n^{j+1}}{(j+1)!} \\ &= n \sum_{j=0}^{\infty} (j+1)^{k-1} \left(e^{-n} \frac{n^j}{j!} \right) = n \mathbb{E}[(X + 1)^{k-1}]. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E}(X^3) &= n^3 + 3n^2 + n, \\ \mathbb{E}(X^4) &= n \mathbb{E}((X + 1)^3) = n [\mathbb{E}(X^3) + 3 \mathbb{E}(X^2) + 3 \mathbb{E} X + 1] \\ &= n(n^3 + 6n^2 + 7n + 1). \end{aligned}$$

Therefore,

$$\mathbb{E}(X - n)^4 = 3n^2 + n$$

and

$$P\left\{X \geq \frac{3}{2}n\right\} \leq \frac{3n^2 + n}{\frac{1}{16}n^4} = \frac{48}{n^2} + \frac{16}{n^3}.$$

(d)

$$P\{X \geq t\} = P\{e^{\theta X} \geq e^{\theta t}\} \leq \frac{\mathbb{E}(e^{\theta X})}{e^{\theta t}} = \frac{e^{n(e^\theta - 1)}}{e^{\theta t}} = e^{-\theta t + n(e^\theta - 1)} = e^{-t \ln \frac{t}{n} + t - n}$$

by taking θ as $\ln(t/n)$.

- Show that $\mathbb{E}(e^{\theta X}) = e^{n(e^\theta - 1)}$.

When $t = \frac{3}{2}n$, the inequality above becomes

$$P\left\{X \geq \frac{3}{2}n\right\} \leq e^{-0.108n}.$$

We see from part (a) to part (d), our tail bound keeps going down, from a constant, first to a rate at $1/n$, then to a rate at $1/n^2$ and finally to an exponential decay in n .

- (3) Flip a fair coin 100 times. By using Central Limit Theorem, estimate the probability of more than 55 heads.

Solution: Let X_j be the result of the j -th flip, so $X_j = 1$ for heads and $X_j = 0$ for tails. The total number of heads is

$$S = X_1 + X_2 + \cdots + X_{100}.$$

As $\mathbb{E}(X_j) = 1/2$, $\text{Var}(X_j) = 1/4$, and $n = 100$, we have

$$\mathbb{E}(S) = 50, \quad \text{Var}(S) = 25, \quad \sigma_S = 5.$$

The central limit theorem says that the standardization of S is approximately $N(0, 1)$. The question asks for $P\{S > 55\}$. Standardizing and using the CLT we get

$$P\{S > 55\} = P\left(\frac{S - 50}{5} > \frac{55.5 - 50}{5}\right) \approx P\{Z > 1.1\} = 0.1357.$$

- (4) If X is a gamma random variable with parameters $(n, 1)$, approximately how large need n be so that

$$\Pr\left\{\left|\frac{X}{n} - 1\right| > 0.01\right\} \leq 0.01?$$

Solution: A $\text{Gamma}(n, 1)$ random variable is the sum of n i.i.d. exponential variables with parameter $\lambda = 1$. Thus $\mathbb{E}(X) = n$ and $\text{Var}(X) = n$. When n is large, by the central limit theorem, $(X - n)/\sqrt{n}$ is approximately a standard normal variable, thus

$$P\left\{\left|\frac{X}{n} - 1\right| > 0.01\right\} \approx P\{|Z| > 0.01\sqrt{n}\}.$$

Looking up the normal CDF table, we see that

$$P\{Z \leq 2.58\} > 0.995,$$

$$P\{Z > 2.58\} < 0.005,$$

$$P\{|Z| > 2.58\} < 0.01,$$

and we need $0.01\sqrt{n} \geq 2.58$, $\sqrt{n} \geq 258$, $n \geq (258)^2$.

- (5) Civil engineers believe that W , the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation 0.3. Approximately how many cars would have to be on the bridge span for the probability of structural damage to exceed 0.1?

Solution: Let $X_1, X_2, \dots$ be the weight of cars, then $\mathbb{E}(X_i) = 3$ and $\text{Var}(X_i) = 0.09$. Let $S_n = X_1 + \dots + X_n$. By the central limit theorem, S_n is approximately $G \sim N(3n, 0.09n)$. On the other hand, $W \sim N(400, 40^2)$. We want to find n such that

$$P\{S_n > W\} \geq 0.1$$

and approximately it is

$$P\{G - W > 0\} \geq 0.1.$$

Note that

$$G - W \sim N(3n - 400, 0.09n + 40^2).$$

Normalizing again,

$$P\{G - W > 0\} = P\left\{Z > \frac{400 - 3n}{\sqrt{0.09n + 40^2}}\right\}$$

Since $P\{Z > 1.28\} \approx 0.1$, we would need

$$1.28 \geq \frac{400 - 3n}{\sqrt{0.09n + 40^2}}$$

from which we can solve that

$$n \geq 116.2,$$

which gives $n \geq 117$.

- (6) A certain component is critical to the operation of an electrical system and must be replaced immediately upon failure. If the mean lifetime of this type of component is 100 hours and its standard deviation is 30 hours, how many of these components must be in stock so that the probability that the system is in continual operation for the next 2000 hours is at least 0.95?

Solution: Let $X_1, X_2, \dots$ be i.i.d $N(100, 30^2)$. We want to find the smallest n such that

$$P\{X_1 + \dots + X_n \geq 2000\} \geq 0.95.$$

Let $S_n = X_1 + \dots + X_n$, then by the central limit theorem, $Z_n = (S_n - 100n)/(30\sqrt{n})$ is approximately $N(0, 1)$ and thus we want

$$P\left\{Z_n \geq \frac{2000 - 100n}{30\sqrt{n}}\right\} \geq 0.95.$$

Since $P\{Z > -1.645\} = P\{Z < 1.645\} \approx 0.95$, we would need

$$-1.645 \geq \frac{2000 - 100n}{30\sqrt{n}},$$

whence we can solve that

$$n \geq 22.33.$$

That is $n \geq 23$.